CTF Writeup: Shadow Identity (OSINT & Forensics)

Target Identification & Digital Evidence Analysis • Archive SHA-256: d5ad9fe2781bd6f933b6c7d1e9480c1ef80f3dab1c55819016be9271c521afae

1. Executive Summary & Solved Flag Candidates

During an intelligence operation, a single photograph (**`miss_grand.jpeg`**) was extracted from a suspect archive (**`shadow-identity.zip`**). Forensic verification confirmed that the JPEG file had all EXIF metadata stripped and contained zero appended bytes or steganographic payloads. Visual intelligence and open-source intelligence (OSINT) analysis identified the subject as **Jihan Mohin Dimack** (also transliterated on her official international contestant profile as **Jihan Mohin Dimachk**, and widely known by her two-part stage name **Jihan Dimack**), who won Miss Grand Tanzania 2026 on August 8, 2026, and subsequently made history on October 10, 2026, as the first African woman crowned Miss Grand International 2026.

Because the challenge prompt specifies a three-token example flag format (**`hiCTF{rahim_ahmed_meow}`**) and requests the person's full name in lowercase, the primary flag uses her three-part full name, accompanied by the two recognized spelling variants below:

Flag Variant Type	Submitted Flag String (Lowercase)	Primary Authority / Source Attribution
Primary 3-Part Full Name (Recommended)	**`hiCTF{jihan_mohin_dimack}`**	The Citizen Tanzania, Indonesian Wikipedia, Swahili Wikipedia (Lead)
Alternate 3-Part Lebanese Transliteration	**`hiCTF{jihan_mohin_dimachk}`**	Xtrafrica (MGI Official Profile), Swahili Wikipedia (Infobox)
2-Part Public Stage Name	**`hiCTF{jihan_dimack}`**	English Wikipedia (MGI 2026), Miss Grand Tanzania

2. Phase 1: Archive Integrity & Binary Forensics Triage

Before conducting OSINT attribution, standard CTF forensic checks were executed on the provided archive **`shadow-identity.zip`** and its extracted payload **`miss_grand.jpeg`** to verify file integrity and rule out hidden steganographic channels or file overlays.

Artifact Filename	Size (Bytes)	File Format & Geometry	Verified SHA-256 Digest
`shadow-identity.zip`	63,218	ZIP Archive (1 file entry)	**`d5ad9fe2781bd6f933b6c7d1e9480c1ef80f3dab1c55819016be9271c521afae`**
`miss_grand.jpeg`	63,052	JPEG / JFIF 1.01, 481×636 px, 8-bit RGB	**`62b55419e589017af5ed17cfd87276f67d7ac355c6b2eec4ae76725239e33a56`**

- **Metadata & EXIF Inspection:** Parsing the JPEG binary structure revealed only standard baseline segments: **`SOI (0xFFD8)`**, **`APP0/JFIF (0xFFE0, 16 bytes)`**, **`DQT (0xFFDB, 132 bytes)`**, **`SOF0 (0xFFC0, 17 bytes)`**, four **`DHT (0xFFC4)`** Huffman tables totaling 190 bytes, and **`SOS (0xFFDA)`** starting at byte offset **`363`**. No **`APP1 (EXIF/XMP)`** or **`COM (0xFFFE)`** comment markers exist.

- **EOF Overlay Check:** The JPEG End of Image marker (`0xFFD9`) resides at byte offset `63,050` out of `63,052` total bytes, confirming exactly **0 trailing bytes** after `EOI`.

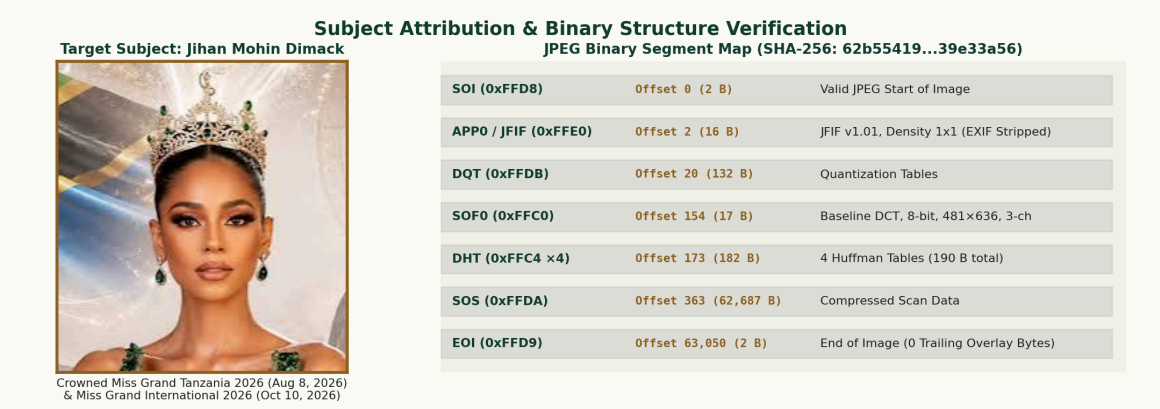


Figure 1: Target subject portrait crop alongside the binary JPEG segment layout of miss_grand.jpeg (confirming stripped EXIF metadata and zero trailing overlay bytes).

3. Phase 2: Visual Intelligence & Landmark Breakdown

Inspecting the visual composition of `miss_grand.jpeg` reveals a promotional pageant poster containing four explicit geographical and institutional clues that pinpoint both the exact title and the specific Tanzanian region represented by the subject:

- **ROI #1 — National Flag of Tanzania (Upper Left):** A waving flag featuring a green upper-left triangle, a light-blue lower-right triangle, and a central black diagonal band fimbriated by yellow stripes, uniquely identifying the United Republic of Tanzania.
- **ROI #2 — Mount Kilimanjaro & Savanna Flora (Middle Left):** The snow-capped volcanic cone of Mount Kilimanjaro rising above an East African umbrella thorn acacia tree on the Serengeti plains.
- **ROI #3 — Bismarck Rock, Mwanza (Middle Right):** The famous precarious granite boulder formation on the shores of Lake Victoria in **Mwanza, Tanzania** ("Rock City"). This landmark serves as a critical corroborating detail: in the national pageant, the subject specifically represented the **Mwanza** region.
- **ROI #4 — Pageant Title Banner (Bottom Center):** Three lines of embossed metallic gold serif typography stating `MISS GRAND TANZANIA 2026` below the contestant wearing an emerald-beaded gown and green gemstone crown.

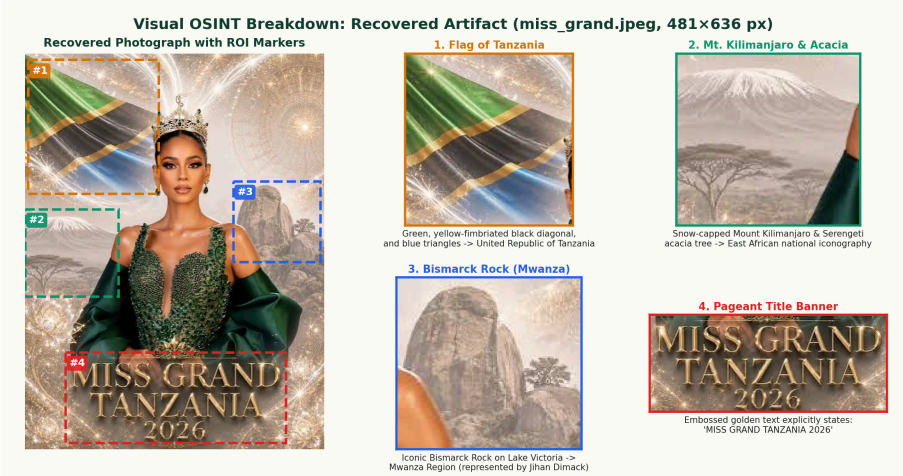


Figure 2: Full recovered photograph (miss_grand.jpeg) annotated with Region of Interest (ROI) markers #1–#4 and corresponding high-resolution evidence crops.

4. Phase 3: Open-Source Intelligence (OSINT) Subject Attribution

Querying open-source records for the winner of Miss Grand Tanzania 2026 who represented the Mwanza region yields an exact, unambiguous match across Tanzanian news outlets and international pageant registries:

Biographical Attribute	Verified Intelligence Finding
Full Legal / Birth Name	**Jihan Mohin Dimack** (also spelled **Jihan Mohin Dimachk**)
Public / Stage Name	**Jihan Dimack**
Date of Birth & Heritage	June 17, 1996 • Tanzanian-Lebanese model and beauty queen (Swahili Wikipedia)
Early Pageant Career	2nd Runner-Up at Miss Tanzania 2014; crowned Miss Universe Tanzania 2016 and represented Tanzania at Miss Universe 2016 in Manila, Philippines
Miss Grand Tanzania 2026	Representing **Mwanza** (matching ROI #3, Bismarck Rock), she was crowned Miss Grand Tanzania 2026 on August 8, 2026, at The Super Dome in Masaki, Dar es Salaam
Miss Grand International 2026	Crowned Miss Grand International 2026 on October 10, 2026, at MGI Hall in Bangkok, Thailand, defeating 76 other delegates to become the first Tanzanian and first African winner in history (The Citizen)

Final Flag Construction

Converting the subject's full name to lowercase and replacing spaces with underscores in accordance with `hiCTF{rahim_ahmed_meow}` yields:

- **3-Part Full Name (Standard Spelling):** `hiCTF{jihan_mohin_dimack}`
- **3-Part Full Name (Lebanese Surname Transliteration):** `hiCTF{jihan_mohin_dimachk}`
- **2-Part Common Name:** `hiCTF{jihan_dimack}`